

# NOAIS - Form 2, Spatial Subtest Documentation

## General information

The Spatial Subtest App is designed for use on desktop web browsers with a mouse. It is not intended for use on mobile devices or touchscreens.

The items are presented in approximately increasing order of difficulty.

After you correctly solve an item, it is removed from the list, and only the remaining unsolved items are displayed.

Each item allows up to three attempts. After three incorrect submissions, the item is marked as failed and can no longer be accessed.

Please note: It is recommended to submit an answer only when you are reasonably confident that it is correct.

## Using the canvas

The interface features a drawing canvas with a 10x10 grid. On the left, a 'Drawing Tools' panel contains buttons for 'Move point' (highlighted in blue), 'Line', 'Arc', 'Matrix', 'Undo', and 'Clear all'. Below the canvas, a text label reads 'Move point mode: drag existing line or arc points only.' At the bottom, there are three buttons: 'Submit' (blue), 'Return to NOAIS-Form 2 index' (white with a blue border), and 'End Test' (blue).

The canvas consists of a  $10 \times 10$  grid of gray squares.

All drawings are composed exclusively of straight lines, circular arcs or matrices.

All points used to create shapes are placed exactly on the intersections of the background grid. The cursor automatically snaps to the nearest grid intersection, making precise placement easier.  
All drawings are displayed in black and white.

## The Toolbox

The toolbox, located to the left of the canvas, contains four drawing tools and two utility buttons.

### Drawing tools

Move Point : Move existing control points.

Line : Draw straight line segments.

Arc : Draw circular arcs.

Matrix : Create matrices.

### Utility buttons

Undo : Reverses the most recent action, you can also use Ctrl + Z. It can be pressed repeatedly to undo multiple actions. It also restores objects removed with Clear All.

Clear All : Removes every object from the canvas. This action can be undone using the Undo button.

### Drawing a line

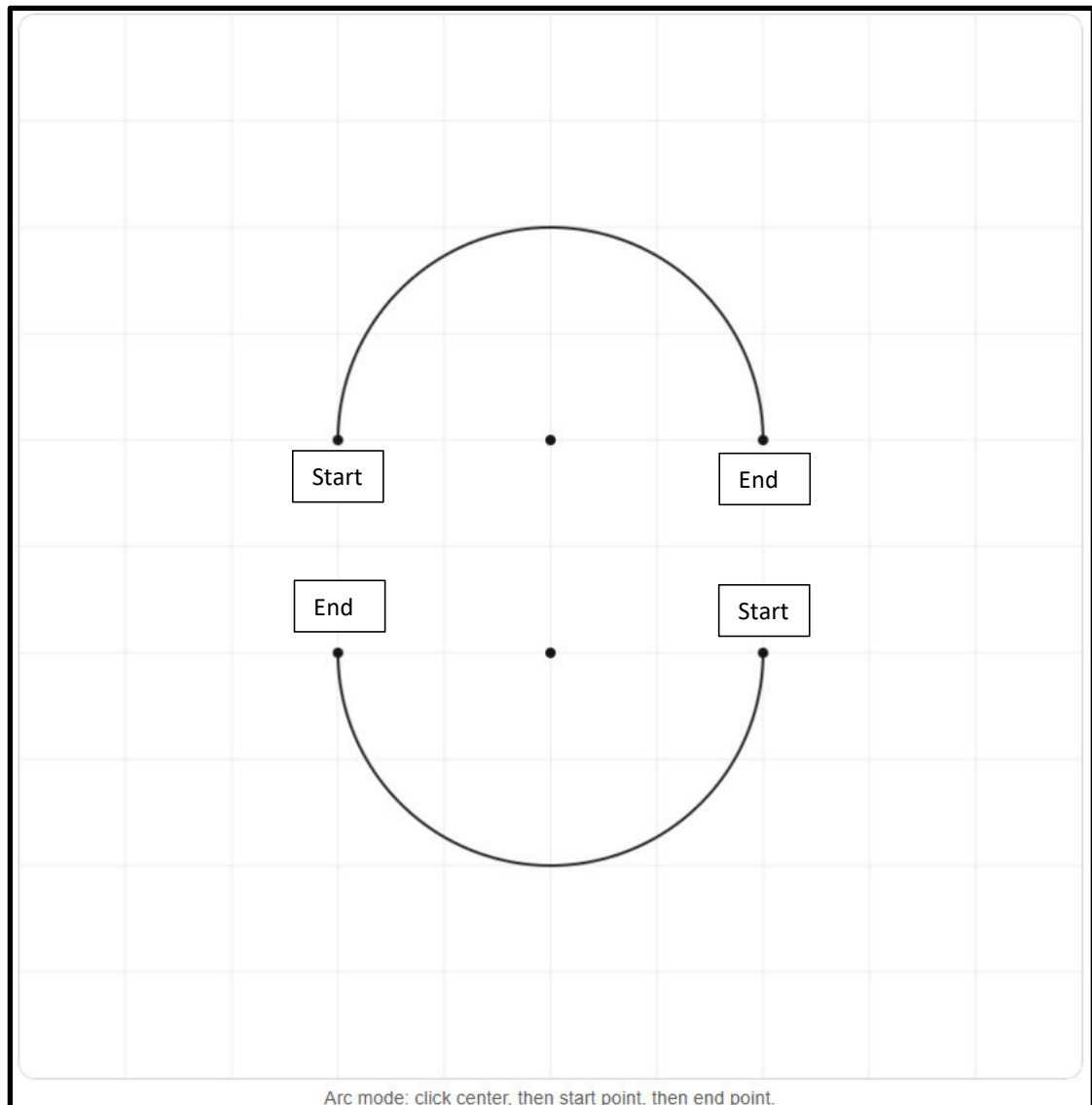
1. Select the Line tool.
2. Click the grid intersection where the line should begin.
3. Click the grid intersection where the line should end.

### Drawing an arc

1. Select the Arc tool.
2. Click the grid intersection corresponding to the center of the circle.
3. Click the point where the arc should begin.
4. Click the point where the arc should end.

Arcs are always drawn clockwise. Therefore, depending on the desired orientation, the starting point must be selected accordingly.

(See the example below.)



## Moving existing shapes

If a point has been placed incorrectly, select the Move Point tool.

Click and drag any control point to a new location.

For circular arcs, moving the center point moves the entire arc while preserving its shape.

## Using Matrix Mode

Selecting the Matrix tool activates Matrix Mode.

While Matrix Mode is active, lines and arcs cannot be drawn.  
Click the Matrix button again to exit Matrix Mode.

## **To create a matrix:**

Click and hold on the grid intersection corresponding to the upper-left corner.  
Drag the mouse until the desired matrix size is reached.  
A live preview is displayed while dragging.

After a matrix has been created:

Drag the bottom-right handle (small black square) to resize it.  
Drag the top-left handle (small black square) to reposition the entire matrix.

To change the color of an individual cell, simply left-click on it.

## **Important note about position**

The absolute position of a drawing on the canvas is not important.  
The scale is not important.  
The number of line/arc segments is not important.

Only the relative proportions between the elements are considered when your answer is checked.

## **Submitting your answer**

Once your drawing is complete, click the Submit button.

Empty canvases cannot be submitted.